

ARIK VAN DER LOO

ELECTRICAL ENGINEERING STUDENT
UNIVERSITY OF WATERLOO - YEAR 4

argon-v.github.io (Portfolio)



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TOOLS

KiCAD AutoCAD & VaultPro STM32 Suite Fusion360 MotorStudio

LANGUAGES

C++ C VHDL Javascript ARMv7 Python

EXPERIENCE

Jan. - Apr.
2025

HARDWARE ENGINEERING INTERN - ALUMAPOWERS - SARNIA & KITCHENER, ON, CANADA

- Designed, assembled, & tested a custom MCU debugger PCB for an STM32 based CPU board.
- Completed hardware bring-up, testing, & validation of power distribution PCB. Identified changes for next version.
- Invested 6 weeks into troubleshooting & tuning a BLDC sensor-less FOC controller for low speed & high torque.
- Wrote application firmware for motor controller & power distribution PCBs using POSIX-like functions in FreeRTOS.
- Created detailed step-by-step documentation for testing, assembly, & firmware updates of 5 different modules.
- Troubleshoot and validated CAN, Ethernet & I2C communication between & within PCBs.
- Researched, sampled, & sourced components. Spoke directly with manufacturers to reduce cost & create relationships.

Major accomplishments:

Successfully tuned motor controller for challenging use case of very slow speed & high startup torque.

May - Aug.
2024

ELECTRICAL ENGINEERING INTERN - RAIN CARBON CANADA - HAMILTON, ON, CANADA

- Worked with electricians and instrumentation technologists in the field to maintain electrical equipment and engineer the seamless integration of new equipment.
- Referenced P&IDs and electrical diagrams to bring 200+ instrumentation loop diagrams up to date and standard.
- Spearheaded engineering design meetings and project walkthroughs with multiple engineering consultants.
- Troubleshoot, diagnosed, and repaired electrical problems with crucial plant process equipment.

Major accomplishments:

Created complete engineering drawing package for a PLC panel in-house; saving the company ~\$30K.

Sep. - Dec.
2023

APPLICATION ENGINEERING CO-OP - SUNGRID SOLUTIONS - CAMBRIDGE, ON, CANADA

- Put together over 75 engineering proposals used to bid on battery energy storage projects across North America.
- Worked together with project managers, engineers, and clients to complete preliminary engineering and CAD design work (including detailed site plans and electrical single line diagrams) for over \$600 million in potential battery energy storage projects.

Major accomplishments:

Co-created a standardised and comprehensive company-wide engineering drawing template.

Jan. - Apr.
2023

EDUCATION

Sep. 2022 -
Present

CANDIDATE FOR: BSC, ELECTRICAL ENGINEERING, HONOURS, CO-OPERATIVE PROGRAM - UNIVERSITY OF WATERLOO - WATERLOO, ON, CANADA

- PCB design course: Assembly and testing of PCB timer board. Design of irrigation system control board.
- 2 Computer logic courses: Introduction to VHDL & assembly programming in ARM
- RF & Microwave Circuits & Radio-Wave Systems courses: Transmission lines & high frequency circuit design & analysis.

PROJECTS

- Power Usage Meter (Group project) -> Design, fabrication, and testing using microchip and LCD display.
- Single Occupant Long Distance Solar Car (Midnight Sun Solar Car Team) -> PCB schematic design using Altium.
- FIRST Robotics 2019 & 2020 challenge robots (High school robotics team) -> Mechanical & Elec. design and fab.